

A Design of Interface for Visual-Impaired People to Access Visual Information from Images Featuring Large Language Models and Visual Language Models

Zhe-Xin Zhang

jacobzhang@digitalnature.slis.tsukuba.ac.jp
Digital Nature Group, University of Tsukuba
Tsukuba, Ibaraki, Japan

Yoichi Ochiai*

Faculty of Library, Information and Media Science,
University of Tsukuba
Tsukuba, Japan
wizard@slis.tsukuba.ac.jp



Figure 1: Setup of the Interface, including graphic tablet, joystick controller, and dot display.

ABSTRACT

We propose a design of interface for visual-impaired People to access visual information from images utilizing Large Language Models (LLMs), Visual Language Models (VLMs), and Segment-Anything. We use Semantic-Segment-Anything to generate the segmentation of semantic objects in images. The segmentation includes two parts: a term set describing the semantic object, and segmented mask which represents the shape of the semantic object. We provide two methods for the visual-impaired user to access the information of the semantic object and its peripheral information in image. In one method, the LLM summarize the term set to create a description. In the other method, the image with the object masked is provided to Visual Language Models which is prompted to respond with a description. In both methods, the mask can be accessed with dot display after processed for the visual-impaired people to access, and the description is prompted to the user in synthesized voice.

*Corresponding author

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648648>

CCS CONCEPTS

• Human-centered computing → User interface programming.

KEYWORDS

Human-Computer Interaction, Large Language Models, Visual Language Models, Segment-Anything

ACM Reference Format:

Zhe-Xin Zhang and Yoichi Ochiai. 2024. A Design of Interface for Visual-Impaired People to Access Visual Information from Images Featuring Large Language Models and Visual Language Models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3613905.3648648>

1 INTRODUCTION

Visual Arts plays a very important role in the field of art. And raising visual arts experiences of visual-impaired people has been a hot topic in the field of human-computer interaction. Franklin Li et al.[4] proposed a research discussing blind people's independent and interdependent access to visual arts utilizing multiple modalities and machine learning. We propose a design of interface featuring Segment-Anything and Visual Language Models for visual-impaired people to access visual information from images.

We used Semantic-Segment-Anything[3] (SSA), an variant implementation for Segment-Anything[2], to generate the segmentation of visual objects in the image. The segmentation includes two parts: the textual description of the visual object, and the segmented mask of the visual object. The description is then handled by Large Language Models(LLMs) to create a description with the visual peripheral information. The mask is then provided to a dot display for the visual-impaired people to access. We also provided an alternative method featuring GPT-4V to directly inspect the image and generate the description. We followed the prompting method of Set-of-Mark proposed by Yang et al.[5]

2 HARDWARE

The hardware interface of our system includes 4 components:

- **Personal Computer:** We used a windows 10 PC to run the software system.
- **Joystick Controller:** Some operations are done by pressing buttons. We chose to use buttons of a Joystick Controller, as it would be easier to distinguish different buttons than keyboard.
- **Graphic Tablet:** We used a Graphic Tablet as mouse to control the cursor. With a graphic tablet whose frame can be touched, the visual-impaired individuals would have a more specific concept of where the cursor is in the whole frame.
- **Dot Display:** We used a dot-figure display as haptic interface for the visual-impaired people to access the shape information of the mask.

2.1 Dot-Figure Display

We used DotView DV-2 from KGS Co., Ltd. as haptic interface for the visual-impaired people to access the shape information of the mask. In touch mode, the screen will show edges of masks, and the dot-figure display will show the edges, which can be touched by the visual-impaired people. DotView DV-2 use its own proprietary software called GView to pick a rectangle area of screen around mouse. The content of that area will than be converted to binary image and shown on the DotView DV-2. The DV-2 has its own algorithm for converting color image to binary image. Some example is shown in Fig.2. It is noticeable that when the cursor moves a short distance, those buttons turned suddenly from raised to concave. This may cause confusion when interacting with semantic objects, so we implemented a workaround to only draw contour, and the information of inside and outside is prompted with Beep sound.

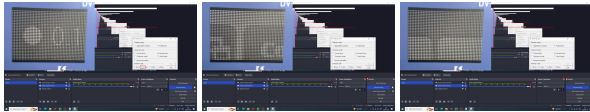


Figure 2: Examples of how DotView DV-2 displays part of screen content. Small red rectangle is the area shown on DV-2. Screenshots took in OBS studio.

3 USER FLOW AND SOFTWARE

3.1 User Flow

The software of interface includes three modes: Full-view mode, Separate mode, and Touch mode.

When using our system, vision-impaired users switch between modes by pressing buttons on the joystick.

We assigned different keys for different modes to avoid confusion. And we use the same key for entering and exiting of each mode, in order to reduce the memory burden for user.

Fig.3 shows how to navigation between modes.

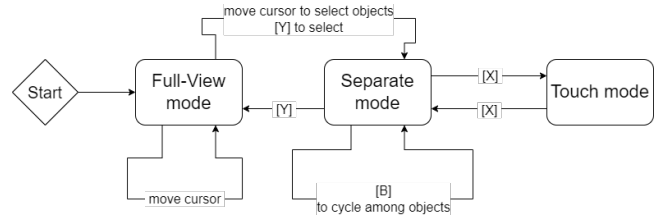


Figure 3: Navigation among modes. [...] means joystick button press, for example, [x] means press on "X" button of joystick

3.1.1 Full-view Mode. The full-view mode is the mode to view the entire image and select a position for further examination. In full-view mode, pressing "Y" button on joystick controller will read out the information of the current position, and then enter Separate Mode, with all the semantic objects at cursor position as "object list" for Separate Mode.

3.1.2 Separate Mode. Sometimes, there can be multiple semantic objects located at a same coordinate in the image. The separate mode is the mode to switch between these objects. In separate mode, pressing "B" button on joystick will select the next object in the aforementioned "object list".

Pressing "X" button on joystick will enter the touch mode with the current object selected. And pressing "Y" on joystick will return to full-view mode. Fig.4 shows an example of selecting two objects at once. The cyan circle indicates the cursor position.

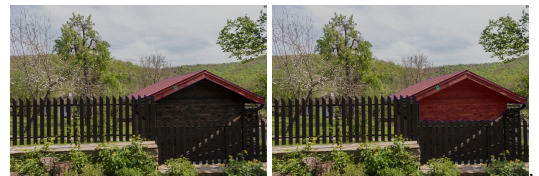


Figure 4: An example of two semantic different objects at same position. Sample image from SSA Official Web Demo(<https://replicate.com/cjwbw/semantic-segment-anything>)

3.1.3 Touch Mode. Touch mode is the mode to access shape information of objects. In touch mode, the screen would show a outline of masks of the semantic object that are selected in separate mode. The Dot Display would show the outline around the cursor, so the visual-impaired user can access the outline by touching the Dot Display. As long as cursor is on the masks, i.e. inside the outline, there would be a Beep sounding continuously. If not, the Beep sound would stop, providing user with the information of inside or outside. Pressing "X" button on joystick will return to separate mode.

3.2 Voice Prompt and Sound Hint

We implemented both voice prompt and sound hint in the system.

We implemented two kinds of sound hint: a light sound hint for "voice prompt not completed" and another relatively heavy sound hint for "process completed". Any input from graphic tablet or joystick controller would be ignored until the "process complete" sound alarms.

The voice prompt would play every time the modes switches, indicating the mode to which user is transitioning into. For example, there will be a voice hint calling out "entering separate mode" when entering Separate mode. This is also applicable for other modes. The description generated by LLM and the labels of objects are also converted into voice prompt for visual-impaired people to access.

3.3 Semantic-Segment-Anything

We used Semantic-Segment-Anything (SSA) to generate a in the format of JSON. Table.1 shows the structure of The JSON object for each object. Fig.5 shows the segmented masks of objects in *Creation of Adam* by Michelangelo. The implementation of Semantic-Segment-Anything we used is from chen et al.[1]. SSA provides not only labels of objects, but also some structural information among objects. For example, if a photo of a man is provided to SSA, the "class_proposals" may include "his hand", rather than only "hand". This vital information is also given to LLM, in order to get a more detailed description.



Figure 5: Example of SSA Segmentation. Left: Original Image(The Creation of Adam, Michelangelo, 1511). Right: Its SSA Segmentation.

Usually, generating segmentation for images using SSA can be very time-consuming, so the segmentation is pre-prepared. Pre-preparing segmentation also allows manually tweaking segmentation, removing the mistakes in segmentation and adding vital information to it. This is especially useful when accessing art with the interface.

3.4 Using LLM and VLM for description

3.4.1 Using LLM. We provide the aforementioned "class_name" and "class_proposals" text-based information to LLM, and let

the LLM summarize the information into a human-understandable description sentence.

For every selected object, we provide not only the information of the object itself, but also the information of surrounding objects. The surrounding objects are defined as objects whose center position is within a range from the original object's center. The center position is the center pixel of "crop_box" for each object.

In our Demo, we use GPT-3.5-turbo as LLM. The LLM is instructed to summarize the information it is given, and convert it into a one-sentence, easy-to-understand description rather than nested lists of key words. The LLM can provide translation while converting, making the interface multilingual.

The advantage of using LLM is that can significantly reduce the process time, making the whole user experience more smooth.

3.4.2 Using VLM. We also implemented a alternative method that uses VLM to describe the selected object and the surrounding of the object. We used GPT-4V(ision) as the VLM in our demo.

In GPT-4V approach, after the object-selection step, a copy of the original image is created, where the object is masked with color and its contour is highlighted. Then GPT-4v is prompted to describe the selected object and the surrounding of the object while provided with the processed image.

The advantage of using GPT-4V is that GPT-4V is able to summarize rich visual context about the whole image content, allowing the user to have a broad-range view of the image.



Figure 6: Example of masking. Left: Example Image, Right: Masked Example Image for GPT-4V.

3.5 Example of user experience

The demo includes a personal computer, and a desk with graphic tablet and pen, joystick controller, and dot display on it. It is optional but strongly recommended to have a normal-sized screen attached to the PC. The screen shows the images real time, and it is vital for assistants.

User should sit in front of the desk, grab the pen of tablet with one hand, and access the dot display and joystick controller with the other hand. The area of tablet surrounded with bezels corresponds to the screen, so that by moving pen on tablet, the user have clue of approximate position of cursor on the screen. The user would then move the cursor among the screen while pressing Y on the joy-con, and explore each semantic object bit by bit in separate mode. If the user is interested in the shape information of a certain semantic object, they can switch into touch mode to access the detailed shape information.

Table 1: The format of result of Semantic-Segment-Anything

Key of JSON dictionary	Type	Content
"segmentation"	dict	Mask saved in COCO RLE format.
"area"	int	The area in pixels of the mask
"bbox"	[x, y, w, h]	The bonding box of the mask
"predicted_iou"	float	The model's own prediction of the mask's quality
"point_coords"	[[x, y]]	The point coordinates input to the model to generate the mask
"stability_score"	float	A measure of the mask's quality
"crop_box"	[x, y, w, h]	The crop of the image used to generate the mask, in XYWH format
"class_name"	str	The most likely category of the object
"class_proposals"	[str]	Top-k most likely categories from Class proposal filter

4 DISCUSSION

Our approach can be used as a method to provide access of visual arts to visual-impaired people, especially photos, portraits or paintings. Segment-Anything models are heavy models and it takes a long time to do the segmentation, makes it nearly not possible to be real-time with current hardware. Also, our interface focused on providing high-resolution information, which also increases the access time of each image for the visual-impaired people. So the expected application scene of this interface is accessing images that are detail-rich, and meaningful enough for the user to spend a lot of time on, like arts, important documents, or photos of precious memorable time.

5 CONCLUSION

We proposed a design of interface for visual-impaired People to access visual information from images using Segment-Anything and LLM. This interface can greatly improve the accessibility of visual information for blind people. With better haptic interface

and manually-labeled segmentation, we expect the visual-impaired people can have a easier access to graphical records in the future.

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A ONLINE RESOURCES

Our demo source code can be accessed on https://github.com/DigitalNatureGroup/sam_cam4blind_demo/